

Student Performance Prediction & Learning Analytics Platform

Data Analytics and Machine Learning Project Report

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This report documents the complete workflow from dataset preparation and exploratory analysis to SQL analysis, machine-learning prediction, and Power BI dashboard development.

1. Introduction

Student academic performance is influenced by several academic, behavioral, family, and learning-related factors. This project develops a data analytics and machine learning workflow to study student performance, identify patterns associated with final grades, analyze students who may require additional academic attention, and present the findings through an interactive Power BI dashboard.

2. Problem Statement

Educational datasets contain useful information about grades, study habits, previous failures, absences, family support, school support, and other student characteristics. The objective is to transform this raw data into meaningful analytical insights and predictive models that can support understanding of student performance.

3. Objectives

The major objectives are to: (1) clean and validate the student dataset, (2) perform exploratory data analysis, (3) analyze relationships between learning factors and final grade, (4) store and query the dataset using MySQL, (5) build machine-learning models for final-grade prediction, (6) identify an at-risk student group using $G3 < 10$, and (7) create an interactive Power BI dashboard for communicating the results.

4. Dataset Description

The project uses the Student Performance dataset obtained from Kaggle. The dataset contains 395 student records and 33 columns. The main academic variables are G1, G2, and G3, where G3 represents the final grade on a 0–20 scale. Other variables include studytime, failures, absences, school, sex, parental education, family support, school support, internet access, and lifestyle-related attributes.

5. Data Cleaning and Preparation

Data cleaning was performed primarily in Microsoft Excel. The dataset was converted into an Excel Table and checked for missing values, duplicates, valid grade ranges, categorical consistency, formatting, and data types. No blank values were found during the missing-value check, and the dataset was reviewed for consistency before analysis. The cleaned workbook was retained in XLSX format so that analysis sheets, PivotTables, and charts could be preserved.

6. Exploratory Data Analysis

Excel and Python were used for exploratory analysis. Key analyses included average G1, G2, and G3; minimum and maximum G3; average final grade by study time, previous failures, gender, school, parental education, internet access, family support, school support, and higher-education aspiration; and an absences-versus-final-grade scatter analysis. Python additionally produced grade distributions and numerical correlation analysis.

7. Key Analytical Observations

The analysis provides evidence of variation in final grades across study-time groups, previous-failure groups, support-related categories, schools, and other student characteristics. Study-time analysis in Excel showed average G3 values of approximately 10.05, 10.17, 11.40, and 11.26 for studytime categories 1 through 4 respectively. The project treats these as descriptive group-level patterns rather

than causal conclusions.

8. SQL Analysis

The dataset was imported into MySQL as the `student_data` table in the `student_performance` database. The SQL workflow verified 395 students and 33 columns and included queries for overall average G3, study-time analysis, previous failures, absences, gender, internet access, parental education, family support, school support, at-risk students, top-performing students, school-level summaries, and performance categories.

9. Machine Learning Methodology

Two Linear Regression models were developed in Python. Model 1 used the available features while retaining G1 and G2 as previous academic performance indicators and used G3 as the target. Model 2 removed G1 and G2 so that prediction focused more on study habits, absences, failures, support variables, and other student characteristics. Categorical variables were encoded using one-hot encoding. The data was divided into training and testing sets using an 80/20 split with `random_state=42`.

10. Model Evaluation

The models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. Model 1 produced MAE = 1.6467, RMSE = 2.3784, and $R^2 = 0.7241$. Model 2 produced MAE = 3.3953, RMSE = 4.1957, and $R^2 = 0.1415$. Lower MAE and RMSE indicate smaller prediction errors, while a higher R^2 indicates that more variance in the test target is explained by the model.

11. At-Risk Student Analysis

For dashboard analysis, students with $G3 < 10$ were treated as an at-risk group. The dataset contains 130 such students. Their average G3 is 5.38, and their average previous-failure value is 0.69. Power BI was used to examine this group by study time, previous failures, school, gender, absences, and individual student characteristics. The threshold is an analytical definition used for this project and should not be interpreted as a formal educational diagnosis.

12. Power BI Dashboard

The Power BI dashboard contains four pages: Overview, Learning Analytics, At-Risk Students Analysis, and ML Prediction. The Overview page presents key KPIs and core charts. Learning Analytics focuses on study habits, support, parental education, and related factors. At-Risk Students Analysis focuses on the $G3 < 10$ group with KPI cards, charts, a student-level table, and slicers. The ML Prediction page presents the evaluation metrics and comparisons of the two machine-learning models.

13. Tools and Technologies

Microsoft Excel was used for data cleaning and spreadsheet-based EDA. Python with pandas, NumPy, Matplotlib, and scikit-learn was used for EDA and machine learning. MySQL and MySQL Workbench were used for database storage and SQL analysis. Power BI Desktop was used to create the final interactive dashboard. VS Code with Jupyter support was used to organize and execute Python notebooks.

14. Project Structure

The final repository is organized into Dataset, Excel, Python, SQL, PowerBI, and Report folders. The Dataset folder contains the source CSV, Excel contains the cleaned analysis workbook, Python contains

the EDA and ML notebooks, SQL contains the SQL analysis script, PowerBI contains the PBIX dashboard, and Report contains this final PDF report.

15. Limitations

The dataset contains 395 records and represents the variables available in the selected source. The linear regression models are baseline predictive models and should not be treated as definitive predictors of student outcomes. Correlation and group-level differences do not establish causation. The at-risk threshold of $G3 < 10$ is a project-defined analytical rule.

16. Future Scope

Future improvements could include testing additional regression algorithms, cross-validation and hyperparameter tuning, richer feature engineering, model explainability, automated data refresh, Power BI service deployment, and integration with a larger or more recent educational dataset. A future system could also incorporate attendance trends and longitudinal academic records where available.

17. Conclusion

This project demonstrates an end-to-end data analytics workflow for student performance analysis. The work combines Excel-based cleaning and analysis, Python-based exploratory analysis and machine learning, MySQL-based data querying, and Power BI-based interactive visualization. The resulting project provides a structured way to explore academic performance, learning-related factors, at-risk students, and predictive model performance.

18. Dataset Source

The project uses the Student Performance Data dataset selected from Kaggle: Devans Odariya, "Student Performance Data." The Kaggle source should be included in the final GitHub README and submission documentation as the original dataset reference.

Final Project Status: Data cleaning, Excel EDA, Python EDA, machine learning, SQL analysis, and four-page Power BI dashboard completed. Final packaging includes the project report, README, and GitHub repository.